

SKILLWEEK10

TASK1: To Register Handlers, Configure Signal Actions, Handle Interrupts, Manage Signal Context, Verify Stability, Test Signal Events

```
GNU nano 7.2 task1.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handle_signal(int sig)
{
    printf("\nSignal received: %d\n", sig);
    printf("Signal handled successfully!\n");
}

int main()
{
    signal(SIGINT, handle_signal);

    printf("Process started.\n");
    printf("PID: %d\n", getpid());
    printf("Press Ctrl+C to send SIGINT.\n");

    while (1)
    {
        printf("Program is running...\n");
        sleep(2);
    }

    return 0;
}
```

```
sanjay_dundra@Sanjay: ~/$ mkdir Skill10
sanjay_dundra@Sanjay: ~/$ cd Skill10
sanjay_dundra@Sanjay: ~/Skill10$ nano task1.c
sanjay_dundra@Sanjay: ~/Skill10$ gcc task1.c -o task1
sanjay_dundra@Sanjay: ~/Skill10$ ./task1
Process started.
PID: 521
Press Ctrl+C to send SIGINT.
Program is running...
Program is running...
Program is running...
^C
Signal received: 2
Signal handled successfully!
Program is running...
Program is running...
Program is running...
^C
Signal received: 2
Signal handled successfully!
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Terminated
sanjay_dundra@Sanjay: ~/Skill10$
```

```
sanjay_dundra@Sanjay: ~/$ cd Skill10
sanjay_dundra@Sanjay: ~/Skill10$ kill -TERM 521
sanjay_dundra@Sanjay: ~/Skill10$
```

TASK2: To Capture SIGINT, Forward Signals, Protect Shell Process, Terminate Foreground Jobs, Update Job States, Test Interrupt Handling

```
sanjay_dundra@Sanjay: ~/ x + v
GNU nano 7.2 task2.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void sigint_handler(int sig)
{
    printf("\nSIGINT received!\n");
    printf("The program received Ctrl+C.\n");
}

int main()
{
    signal(SIGINT, sigint_handler);

    printf("Program PID: %d\n", getpid());
    printf("Press Ctrl+C to test SIGINT.\n");

    while (1)
    {
        printf("Foreground job is running...\n");
        sleep(3);
    }

    return 0;
}
```

```
sanjay_dundra@Sanjay:~/Skill10$ nano task2.c
sanjay_dundra@Sanjay:~/Skill10$ gcc task2.c -o task2
sanjay_dundra@Sanjay:~/Skill10$ ./task2
Program PID: 566
Press Ctrl+C to test SIGINT.
Foreground job is running...
Foreground job is running...
^C
SIGINT received!
The program received Ctrl+C.
Foreground job is running...
Foreground job is running...
Foreground job is running...
^C
SIGINT received!
The program received Ctrl+C.
Foreground job is running...
^Z
[1]+  Stopped                  ./task2
sanjay_dundra@Sanjay:~/Skill10$ jobs
[1]+  Stopped                  ./task2
sanjay_dundra@Sanjay:~/Skill10$ bg %1
[1]+ ./task2 &
Foreground job is running...
sanjay_dundra@Sanjay:~/Skill10$ jForeground job is running.
..
obs
[1]+  Running                  ./task2 &
sanjay_dundra@Sanjay:~/Skill10$ Foreground job is running...
.
Foreground job is running...
```

```

sanjay_dundra@Sanjay:~/Skill10$ Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
^C
sanjay_dundra@Sanjay:~/Skill10$ Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
^C
[1]+  Terminated                  ./task2
sanjay_dundra@Sanjay:~/Skill10$ █

```

TASK3: To Capture SIGTSTP, Suspend Processes, Update Job Table, Preserve State, Resume Later, Test Suspension Workflow

```

GNU nano 7.2 task3.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void sigtstp_handler(int sig)
{
    printf("\nSIGTSTP received!\n");
    printf("Process received Ctrl+Z.\n");
}

int main()
{
    signal(SIGTSTP, sigtstp_handler);

    printf("Process PID: %d\n", getpid());
    printf("Press Ctrl+Z to send SIGTSTP.\n");

    while (1)
    {
        printf("Process is running...\n");
        sleep(2);
    }

    return 0;
}

```

```

sanjay_dundra@Sanjay:~/Skill10$ nano task3.c
sanjay_dundra@Sanjay:~/Skill10$ gcc task3.c -o task3
sanjay_dundra@Sanjay:~/Skill10$ ./task3
Process PID: 706
Press Ctrl+Z to send SIGTSTP.
Process is running...
Process is running...
Process is running...
^Z
SIGTSTP received!
Process received Ctrl+Z.
Process is running...
Process is running...
Process is running...
Process is running...
Process is running...
Process is running...
^Z
SIGTSTP received!
Process received Ctrl+Z.
Process is running...
Process is running...
Process is running...
Process is running...
Process is running...
Process is running...
Terminated

```

TASK4: To Create Process Groups, Assign Group IDs, Transfer Terminal Control, Restore Ownership, Coordinate Signals, Validate Behavior

```

GNU nano 7.2 task4.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main()
{
    pid_t pid = getpid();
    pid_t ppid = getppid();
    pid_t pgid = getpgrp();
    pid_t sid = getsid(0);

    printf("Process ID (PID): %d\n", pid);
    printf("Parent PID: %d\n", ppid);
    printf("Process Group ID (PGID): %d\n", pgid);
    printf("Session ID (SID): %d\n", sid);

    printf("\nProcess is running...\n");

    while (1)
    {
        sleep(2);
    }

    return 0;
}

```



```
sanjay_dundra@Sanjay: ~/ x + v - □ x
sanjay_dundra@Sanjay:~/Skill10$ nano task4.c
sanjay_dundra@Sanjay:~/Skill10$ gcc task4.c -o task4
sanjay_dundra@Sanjay:~/Skill10$ ./task4
Process ID (PID): 1180
Parent PID: 388
Process Group ID (PGID): 1180
Session ID (SID): 388

Process is running...
^C
sanjay_dundra@Sanjay:~/Skill10$ ps -o pid,ppid,pgid,sid,tpgid,stat,cmd
  PID   PPID   PGID   SID   TPGID  STAT  CMD
   388    387    388    388    1181  Ss    -bash
  1160    388   1160    388    1181  S     sleep 100
  1181    388   1181    388    1181  R+    ps -o pid,ppid,pgid
sanjay_dundra@Sanjay:~/Skill10$ kill -TERM -1180
-bash: kill: (-1180) - No such process
[1]+  Done                  sleep 100
sanjay_dundra@Sanjay:~/Skill10$ kill -TERM -1160
-bash: kill: (-1160) - No such process
sanjay_dundra@Sanjay:~/Skill10$ ./task4 &
[1] 1191
sanjay_dundra@Sanjay:~/Skill10$ Process ID (PID): 1191
Parent PID: 388
Process Group ID (PGID): 1191
Session ID (SID): 388

Process is running...
sanjay_dundra@Sanjay:~/Skill10$ jobs -l
[1]+  1191 Running          ./task4 &
sanjay_dundra@Sanjay:~/Skill10$ kill -STOP 1191
sanjay_dundra@Sanjay:~/Skill10$ jobs
[1]+  Stopped              ./task4
sanjay_dundra@Sanjay:~/Skill10$ kill -CONT 1191
sanjay_dundra@Sanjay:~/Skill10$ jobs
[1]+  Running              ./task4 &
sanjay_dundra@Sanjay:~/Skill10$ |
```


